P3 - Scalable Lakehouse Pipeline

Portfolio Project 3 | PySpark + Parquet + Delta + Fabric | RC2

Business scenario

A logistics network produces large shipment-event files and reference data. Build a scalable Bronze/Silver/Gold lakehouse pipeline using PySpark, Parquet and Delta, then document how it runs in Microsoft Fabric. The design must handle schema problems, incremental MERGE, partition/performance decisions and quality checks.

Required deliverables

- Define explicit schemas and reject malformed events.
- Build Bronze raw, Silver cleaned/latest-state and Gold business outputs.
- Join reference data and use window logic correctly.
- Choose Parquet/Delta partitioning based on workload, not unique IDs.
- Implement Delta MERGE/upsert and prove logical idempotency.
- Inspect/explain shuffle, partition count, small files and one performance optimization.
- Define Fabric OneLake/Lakehouse/notebook/pipeline deployment and monitoring.
- Optionally add streaming only when justified by the business requirement.
- Trigger and repair supplied schema/skew/small-file failures.
- Deliver README, architecture, runbook, performance notes and defense.

Deliberate failure pack

- Wrong inferred schema
- Malformed event
- Skewed join key
- Too many small files
- Unique-ID partition anti-pattern
- Incremental update
- Schema evolution
- Duplicate update
- Unnecessary cache

Submission rule

Run the local project checker, preserve failure/recovery evidence, and save a final README + architecture + runbook. Live tool/runtime requirements remain conditional where the build environment cannot execute them.